Readers make targeted regressions to likely word errors in noisy-channel garden paths

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Background. A key question in psycholinguistics is how readers perform **reanalysis** during incremental processing. Unlike auditory processing, where later context can influence word identification [2], reading exposes overt reanalytical behavior. The Selective Reanalysis Hypothesis [4] states that readers target locations which call for syntactic reanalysis during re-reading; regressive eye movements may also help with resolving uncertainty about visual input [1]. However, it remains unknown whether selective reanalysis extends to *error-sensitive* reanalysis of previous material (e.g. deciding that a word was an error in light of later-arriving, surprising information), as opposed to syntactic reanalysis. A recent information-theoretic analysis of naturalistic reading data showed that regressions are more common between pairs of words with *high*, not low, mutual information, supporting a **reactivation** account of regressions over a reanalysis one [7]. Meanwhile, a noisy-channel model [3,5,6] predicts that readers perform probabilistic inference to identify errors and propose alternative interpretations. In this work, we study the dynamics of reading for "noisy-channel garden path" (GP) sentences like *"The boy **licked** the big round ball into the net"*, which temporarily appear well-formed, but which have late-appearing violations of expectation at the *predicate* that can be resolved by inferring a production error at an earlier ***critical word***, which has a plausible alternative as an orthographic neighbor. We compare against minimally different plausible sentences (*kicked*…*ball*) and typo errors which do not induce a garden path (*kjcked*…*ball*). ***Materials and Methods.*** We design N=36 minimally differing items belonging to the conditions of Plausible, Gardenpath-Neighbor, and Typo-Error, each consisting of a preamble, critical word, intervening region, and predicate (**Table 1**). The **targeted regression account** predicts that readers will tend to initiate regressions towards the critical word upon encountering an implausible predicate (i.e. a region–condition interaction in the GP-Neighbor and Typo-Error conditions). Alternatively, readers may simply re-read sentences either in full or by sampling randomly (non-targeted accounts), or engage in targeted regressions to non-erroneous parts of sentences (reactivation). We use the Mouse Tracking for Reading paradigm [8] to gather word-level reading measures from N=200 participants via Prolific. Participants also reported whether or not each sentence contained an error. We compare reading measures across conditions and regions (**Figure 1**), and additionally use a hurdle-lognormal mixed-effects regression to predict per-word re-reading duration (total minus gaze duration), from the interaction of condition and region, with control predictors of surprisal, word position, word length, frequency, and part of speech. ***Results and Discussion.*** Participants systematically distinguished plausibility, reporting errors in 11% of plausible, 62% of GP-Neighbor, and 89% of Typo-Error sentences. For typos, readers often stop reading after detecting the obvious error. Our results show **evidence for targeted reading regressions** towards words that are likely errors and therefore promising loci for reanalysis, in light of later-arriving information. In particular, in the GP-Neighbor condition, readers are more likely to initiate regressions at the predicate than at other regions or in other conditions, suggesting an attempt at reanalysis in response to the implausible predicate; meanwhile, the rate of regressions-into and re-reading duration at the critical word were both significantly higher in GP-neighbor than in other conditions (**Figure 1**; statistical significance denoted by asterisks). There are significant region–condition interactions for GP-Neighbor and Typo-Error at the critical word (see **Table 2** for statistical model results), implying that regressions are targeted rather than uniform. While the Typo condition sees relatively few regressions, when present, regressions are highly targeted to the location of the typo. The results contrast with **a)** non-targeted accounts of regressions, where anomalies trigger content-agnostic re-reading and **b)** strict reactivation accounts of regressions – in a direct manipulation, regressions are more likely between pairs of words which are semantically incongruous, rather than mutually predictive. These results shed light on the incremental dynamics of how readers allocate their gaze to recover from errors in language input.

References

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Condition	Preamble	Critical	Intervening	Predicate	Description
Plausible	The boy	kicked	the big round	ball into the net.	Sentence is semantically plausible.
	The boy	licked	the big round	lollipop with delight.	
GP-neighbor	The boy	licked	the big round	ball into the net.	Temporarily plausible until predicate; critical word is neighbor of plausible word.
	The boy	kicked	the big round	lollipop with delight.	
Typo-Error	The boy	kjcked	the big round	ball into the net.	Critical word is a nonword neighbor of plausible word; error immediately apparent.
	The boy	ljcked	the big round	lollipop with delight.	

Table 1: Experimental item example. Items are counterbalanced across Plausible and GP-neighbor sentences such that the same word appears as plausible and implausible depending on the rest of the sentence. The critical word in both GP-neighbor and Typo-Error differ by one edit from a plausible counterpart, while GP-neighbor but not Typo-Error induces a garden path effect due to being temporarily plausible.

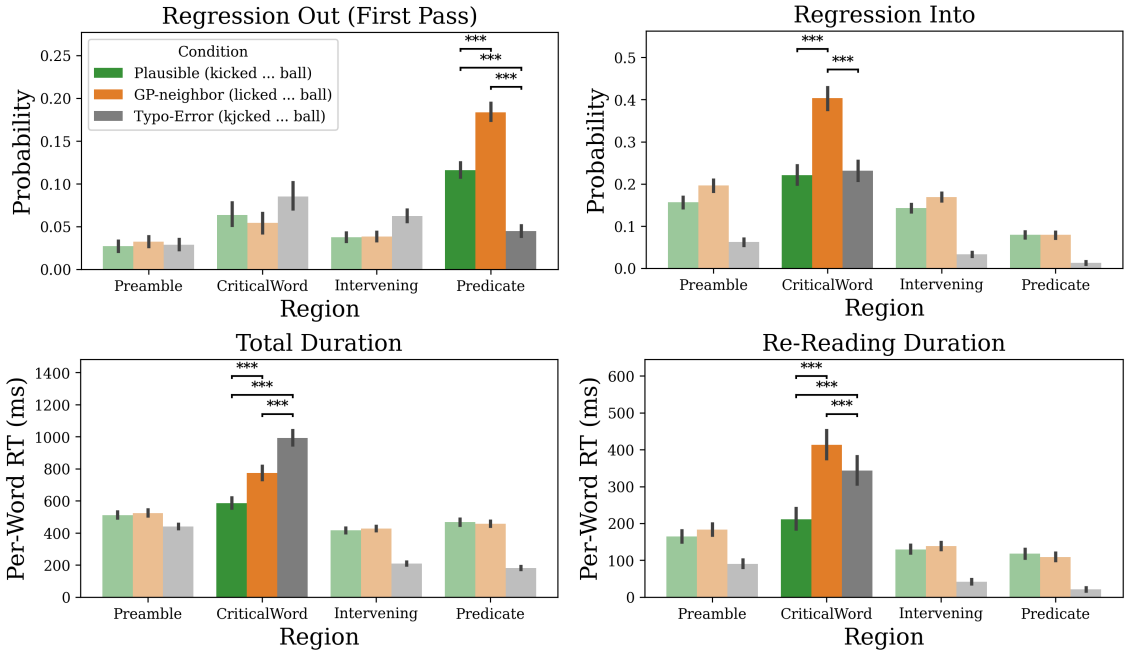

Figure 1: Reading Measures by region and condition, with 95% confidence intervals. Asterisks denote significance level (***: $p < 0.001$; **: $p < 0.01$; *: $p < 0.05$; no label: not significant) of Mann-Whitney U-test between condition pairs at the region of interest (*Predicate* for regressions-out, *CriticalWord* for all other measures).

Measure	Predictor	Estimate (β)	Credible Interval	P(β > 0 | data)
Re-Reading Duration	GP-neighbor:CriticalWord	0.21	[0.12, 0.31]	1.0000
Re-Reading Duration	Typo-Error:CriticalWord	0.46	[0.33, 0.58]	1.0000

Table 2: Estimates of interactions in the lognormal component of fitted mixed-effects hurdle-lognormal model, with Plausible as the reference condition and Intervening as the reference region. Estimates are in log space: e.g. **β =** 0.21 implies a multiplicative factor of exp(0.21) = 1.23. For both of the above interactions, the CrI does not include 0.